re

Map_Maze

```
  *(_DWORD *)v6[i24 + 75] = 1;
*(_DWORD *)v6[84] = 0;
*(_DWORD *)v6[85] = 1;
*(_DWORD *)v6[86] = 1;
*(_DWORD *)v6[87] = 0;
for ( i25 = 13; i25 < 15; ++i25 )
  *(_DWORD *)v6[i25 + 75] = 1;
for ( i26 = 0; i26 < 9; ++i26 )
  *(_DWORD *)v6[i26 + 90] = 1;
*(_DWORD *)v6[99] = 0;
*(_DWORD *)v6[100] = 1;
*(_DWORD *)v6[101] = 1;
*(_DWORD *)v6[102] = 0;
for ( i27 = 13; i27 < 15; ++i27 )
  *(_DWORD *)v6[i27 + 90] = 1;
for ( i28 = 0; i28 < 12; ++i28 )
  *(_DWORD *)v6[i28 + 105] = 1;
for ( i29 = 0; i29 < 15; ++i29 )
{
  for ( i30 = 0; i30 < 15; ++i30 )
  {
    if ( i29 > 0 )
      *(_DWORD *)(v5[15 * i29 + i30] + 4) = v5[15 * i29 - 15 + i30];
    if ( i29 < 14 )
      *(_DWORD *)(v5[15 * i29 + i30] + 8) = v5[15 * i29 + 15 + i30];
    if ( i30 > 0 )
      *(_DWORD *)(v5[15 * i29 + i30] + 12) = v5[15 * i29 - 1 + i30];
    if ( i30 < 14 )
      *(_DWORD *)(v5[15 * i29 + i30] + 16) = v5[15 * i29 + 1 + i30];
  }
}
v2 = (_DWORD *)v5[0];
00000CB7 sub_4010D0:124 (4018B7)
```

这里对迷宫做了初始化，以及定义了上下左右的移动方式

根据初始化的结果得到迷宫的形状

0,1,1,1,1,1,1,1,1,1,1,1,1,1,1,

0,0,0,0,0,0,0,0,0,1,1,1,1,1,1,

1,1,0,1,1,1,1,1,0,1,1,1,1,1,1,

1,1,0,1,1,1,1,1,0,0,0,0,1,1,1,

1,1,0,0,0,0,0,1,0,0,0,1,1,1,1,

1,1,0,1,1,1,0,0,0,0,0,1,1,1,1,

1,1,0,1,1,1,0,1,1,1,0,1,1,1,1,

1,0,0,1,1,1,0,1,1,1,0,1,1,1,1,

1,1,0,0,0,0,0,1,1,1,0,1,1,1,1,

1,1,1,1,1,1,0,1,1,1,0,1,1,1,1,

1,1,1,1,1,1,0,0,0,0,0,1,1,1,1,

1,1,1,1,1,1,1,1,1,0,0,0,0,1,1,

1,1,1,1,1,1,1,1,1,0,1,1,0,1,1,

1,1,1,1,1,1,1,1,1,0,1,1,0,1,1,

1,1,1,1,1,1,1,1,1,1,1,1,0,0,0

获得上下左右对应的字符UDLR

```
char __cdecl sub_401C40(_DWORD *a1, _DWORD *a2, int a3)
{
  char result; // al
  int i; // [esp+0h] [ebp-8h]

  for ( i = 0; *(_BYTE *)(i + a3); ++i )
  {
    if ( *(_BYTE *)(i + a3) == 'U' && a1[1] && !*(_DWORD *)a1[1] )
    {
      a1 = (_DWORD *)a1[1];
    }
    else if ( *(_BYTE *)(i + a3) == 'D' && a1[2] && !*(_DWORD *)a1[2] )
    {
      a1 = (_DWORD *)a1[2];
    }
    else if ( *(_BYTE *)(i + a3) == 'L' && a1[3] && !*(_DWORD *)a1[3] )
    {
      a1 = (_DWORD *)a1[3];
    }
    else
    {
      if ( *(_BYTE *)(i + a3) != 'R' || !a1[4] || *(_DWORD *)a1[4] )
        return 0;
      a1 = (_DWORD *)a1[4];
    }
  }
  result = (char)a2;
  if ( a1[2] == a2[2] )
  {
    result = (char)a2;
    if ( a1[1] == a2[1] )
    {
00001102 sub_401C40:22 (401D02)
```

根据迷宫形状，猜测0，0位置为起点，15，15位置为终点，写一个bfs的脚本

```
from collections import deque


maze = [
    [0,1,1,1,1,1,1,1,1,1,1,1,1,1,1],
[0,0,0,0,0,0,0,0,0,1,1,1,1,1,1],
[1,1,0,1,1,1,1,1,0,1,1,1,1,1,1],
[1,1,0,1,1,1,1,1,0,0,0,0,1,1,1],
[1,1,0,0,0,0,0,1,0,0,0,1,1,1,1],
```

```
[1,1,0,1,1,1,0,0,0,0,0,1,1,1,1],
[1,1,0,1,1,1,0,1,1,1,0,1,1,1,1],
[1,0,0,1,1,1,0,1,1,1,0,1,1,1,1],
[1,1,0,0,0,0,0,1,1,1,0,1,1,1,1],
[1,1,1,1,1,1,0,1,1,1,0,1,1,1,1],
[1,1,1,1,1,1,0,0,0,0,0,1,1,1,1],
[1,1,1,1,1,1,1,1,1,0,0,0,0,1,1],
[1,1,1,1,1,1,1,1,1,0,1,1,0,1,1],
[1,1,1,1,1,1,1,1,1,0,1,1,0,1,1],
[1,1,1,1,1,1,1,1,1,1,1,1,0,0,0],
]


directions = [(-1, 0, 'U'), (1, 0, 'D'), (0, -1, 'L'), (0, 1, 'R')]


def in_bounds(x, y):
    return 0 ≤ x < 15 and 0 ≤ y < 15


def bfs(maze):
    queue = deque([((0, 0), [(0, 0)], 0, '')])
    visited = set([(0, 0)])
    valid_paths = []

    while queue:
        (x, y), path, length, directions_path = queue.popleft()

        if (x, y) == (14, 14) and length == 28:
            valid_paths.append(directions_path)

        for dx, dy, dir_char in directions:
            nx, ny = x + dx, y + dy

```

```
                if in_bounds(nx, ny) and maze[nx][ny] == 0 and (nx, ny) not in visited:
                    visited.add((nx, ny))
                    queue.append(((nx, ny), path + [(nx, ny)], length + 1, directions_path + dir_char))

    return valid_paths


paths = bfs(maze)
if paths:
    print(f"找到 {len(paths)} 条路径: ")
    for path in paths:
        print(path)
else:
    print("没有找到符合条件的路径")
```

得到结果

DRRDDDDDDDRRRRDDRRRDRRRDDDRR

md5得到flag

LZSDS{1979869e0c4ef6c542e54ae5c48f63ec}

superpandagirl

用dnSpy打开Super Panda Girl_Data->Managed->Assembly-CSarp.dll

找到获取flag的函数VerifyPassword()，这段函数在验证了输入的text和经过pandaKey加密后的text相同后，通过GetEvenCharacters()函数选取text中下标为偶数的字符组成新字符串，再通过code()函数进行base64加密

对pandaKey的s5Ss55s5s5s5sSs5类进行分析，发现是经过修改（最后没有array[b3]，改为直接b3）的RC4加密，key为 `LZSDS` ，加密前的字节为 `57, 244, 117, 200, 213, 87, 194, 195, 164, 100, 103, 63, 19, 79, 137, 70, 201, 24, 163, 129, 237, 210, 5, 19, 35, 21`

写出对应的脚本：

```
import base64
def rc4(key, ciphertext):
    # 初始化 S-box
    S = list(range(256))
    key_len = len(key)
    # KSA
    j = 0
    for i in range(256):
        j = (j + S[i] + key[i % key_len]) % 256
        S[i], S[j] = S[j], S[i]
    # PRGA
    i = j = 0
    plaintext = []
    for byte in ciphertext:
        i = (i + 1) % 256
        j = (j + S[i]) % 256
        S[i], S[j] = S[j], S[i]
        K = (S[i] + S[j]) % 256
        plaintext.append(byte ^ K)
    return bytes(plaintext)

key = b"LZSDS"
ciphertext = bytes([57, 244, 117, 200, 213, 87, 194, 195, 164, 100, 103, 63, 19, 79, 137, 70, 201, 24, 163, 129, 237, 210, 5, 19, 35, 21])
decrypted = rc4(key, ciphertext)  #rc4
text = decrypted.decode('utf-8')
```

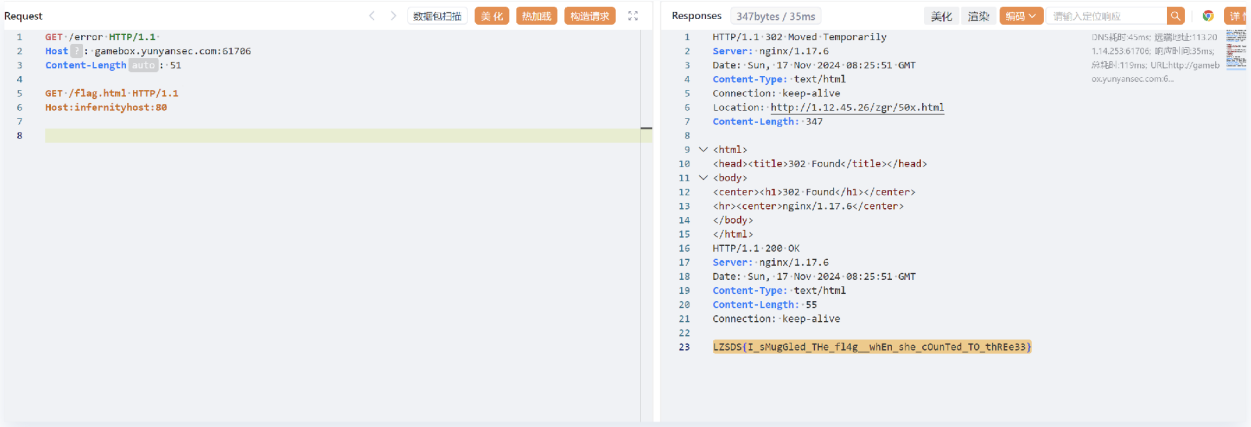
```
binary_data = text[::2].encode("utf-8") #选取下标为偶数位
str = base64.b64encode(binary_data) #to base64
print(str)
```

得到最终的字符串 cHR0aV9udGV0dV9ybg== ，按ShowCorrectPasswordUI("LZSDS{" + str + "}")补充得到flag

web

海关警察训练平台

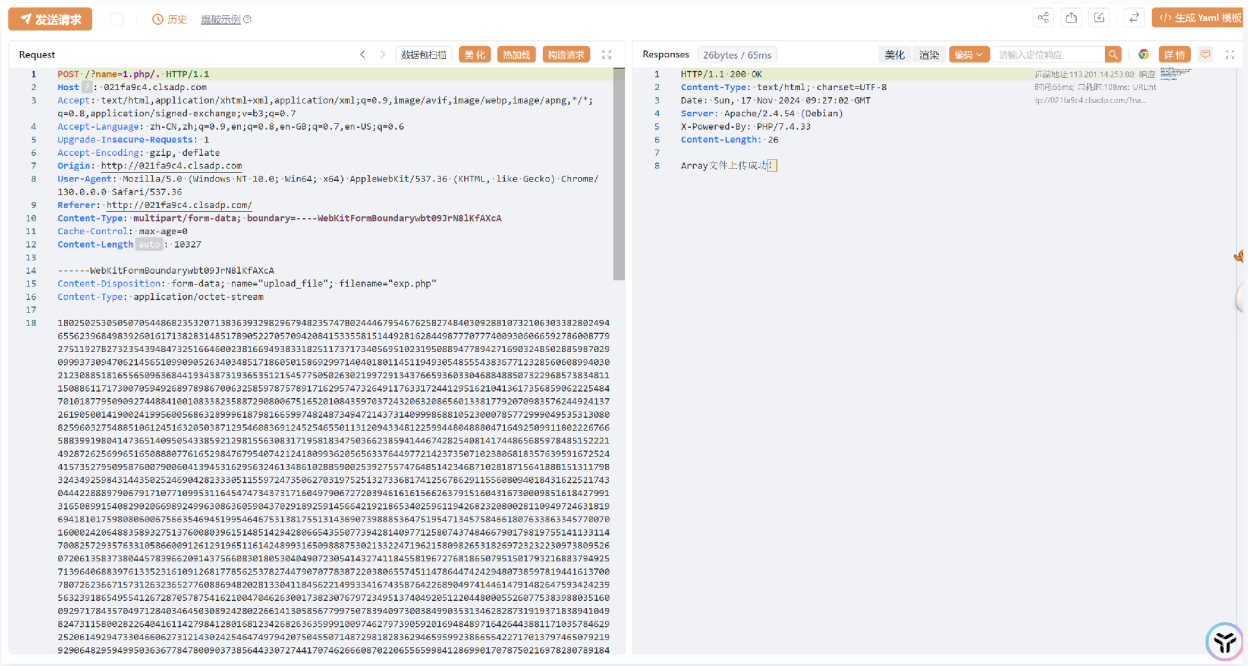
查看js源码发现全部判断正确会访问/error，访问一下发现重定向了，以及相应包有返回nginx的版本1.17.6，搜索相关漏洞搜到了https://www.cnblogs.com/null1433/p/12778026.html



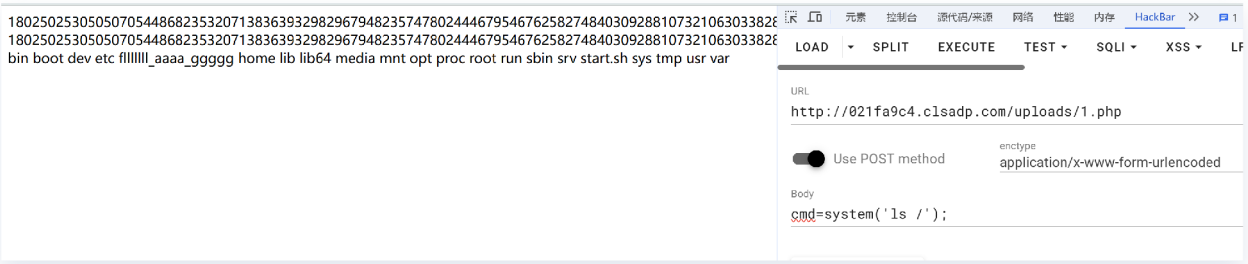
奶龙牌WAF

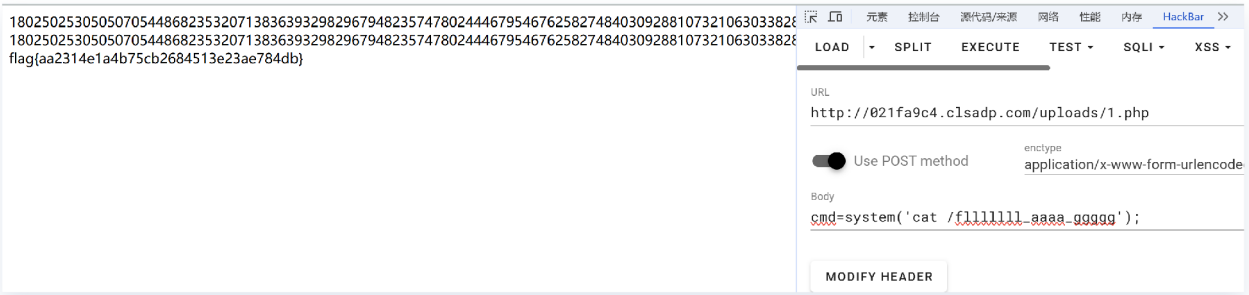
查看源码，发现对文件类型的检测用的是pathinfo以及上传文件是move_uploaded_file，可以使用/.绕过

waf只对前5000字节检测，所以只需要生成5000字节的乱码再后面加代码就行



文件末尾加上





crypto

xorsa

真签到，gcd(e,phi)==2，直接开根就行了

```python
from Crypto.Util.number import *
from gmpy2 import *
import libnum

c = 13760578729891127041098229431259961120216468948795732373975536417751222443069805775693845560005881981622202089883866395577154701229046245882282127054969114210307175116574178428823043817041956207503299220721042136515863979655578210499512044917781566303947681251248645504273995402630701480590505840473412765662
n = 14247038211821385209759067256846232227444163173099199085257790370590450749665206556163364754269182255358084948354345827898987234756662133974633117062902370811855466665351784027125333112663075085395676501121759786699720149098576433141817737564928779420725539793335830274229206316999461309927000523188222801659
hint1 = 8938538619961731399716016665470564084986243880394928918482374295814509353382364651201249532111268951793354572124324033902502588541297713297622432670722730
q = 14932981552434748373200928493257503877595196438793886092083144940006055540206367063208490329067591219147624923784898525755832601775465789353209776130506 47

p = n // q
assert p^q == hint1

e=2026
phi = (p-1) * (q-1)

d = gmpy2.invert(e//2,phi)

m1 = pow(c,d,n)
```

```
m = gmpy2.iroot(m1,2)
flag = libnum.n2s(int(m[0]))
print(flag)
```

misc

欢迎来到2024蜀道山CTF

公众号关注就行

神奇的硬币纺纱机

nc连过去之后，为了拿到100个硬币，只要不投就能稳定拿对面的硬币，全选0就可以

Elemental Wars

五行，一开始以为只要一直选一个就可以，后来发现不太行。然后选择了一直选择对面本回合的属性，玩就完事了

golf

简单的一个python的逃逸

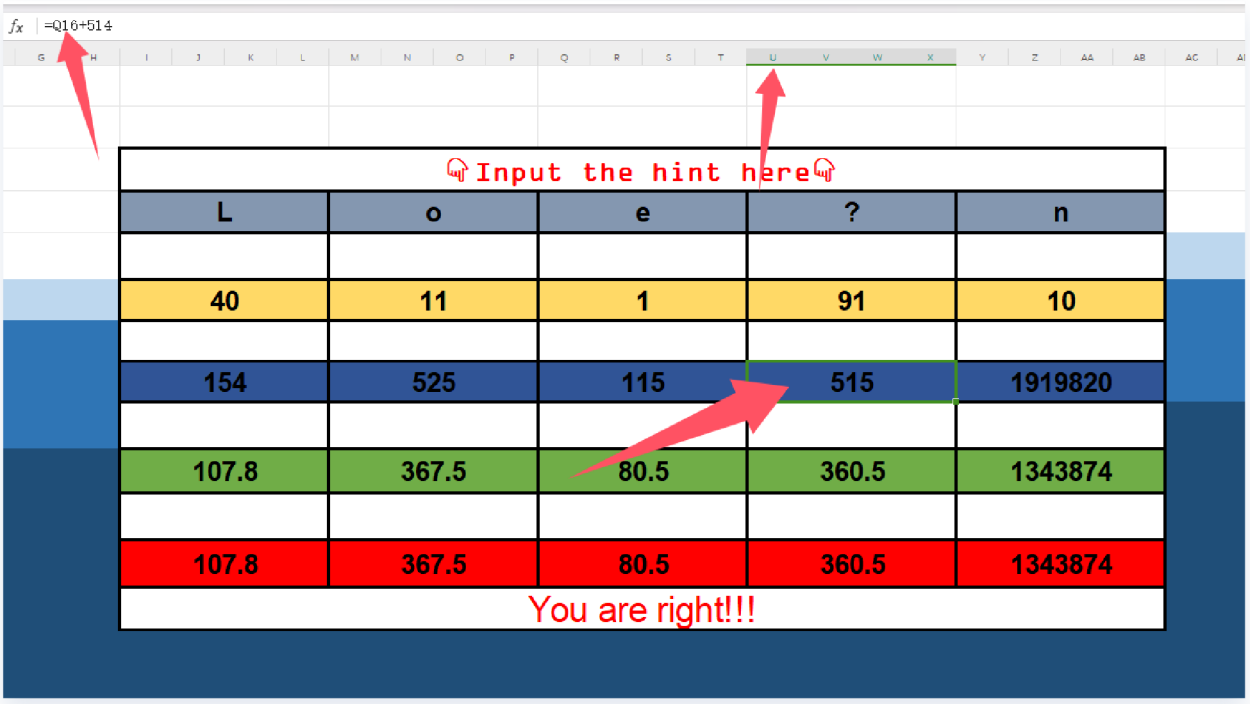
把要执行的命令base64加密一下就可以，长度不能超过187，不要出现黑名单的内容即可：

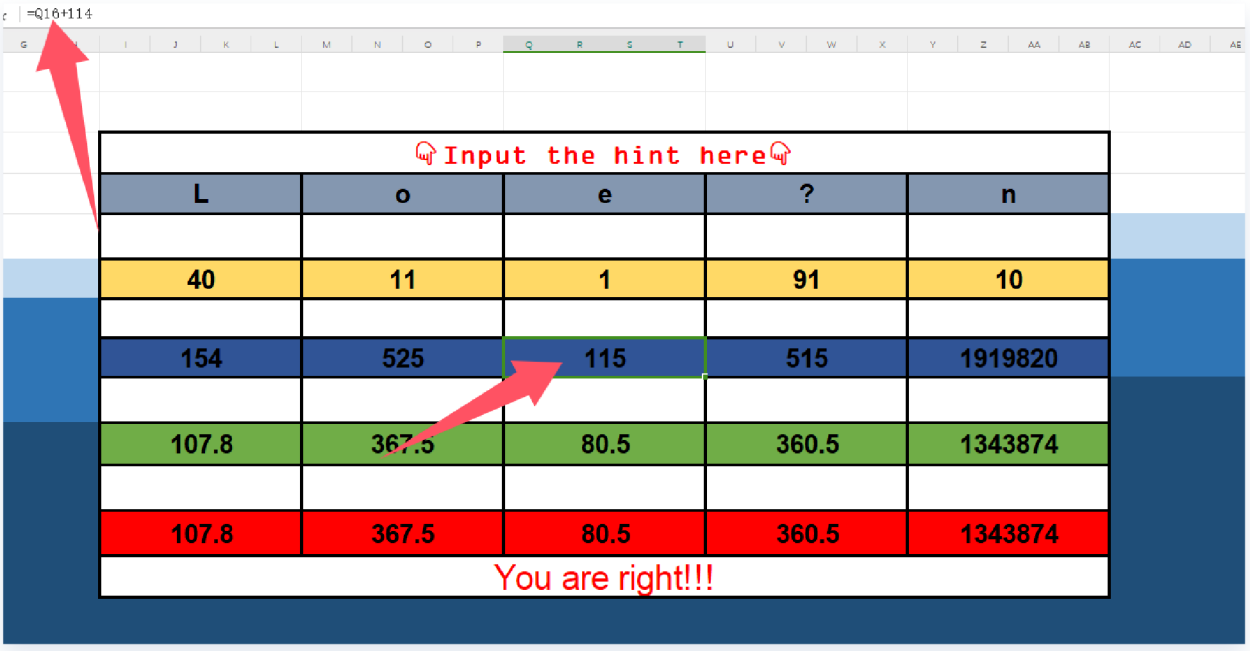
此处为了让s == BOX就行，BOX被ban了，那就用全角字符就行了s=ＢＯＸ，base64一下

cz3vvKLvvK/vvLgK

Summit Potato

首先png尾藏了一个xlsx，点进sheet2完成闯关：

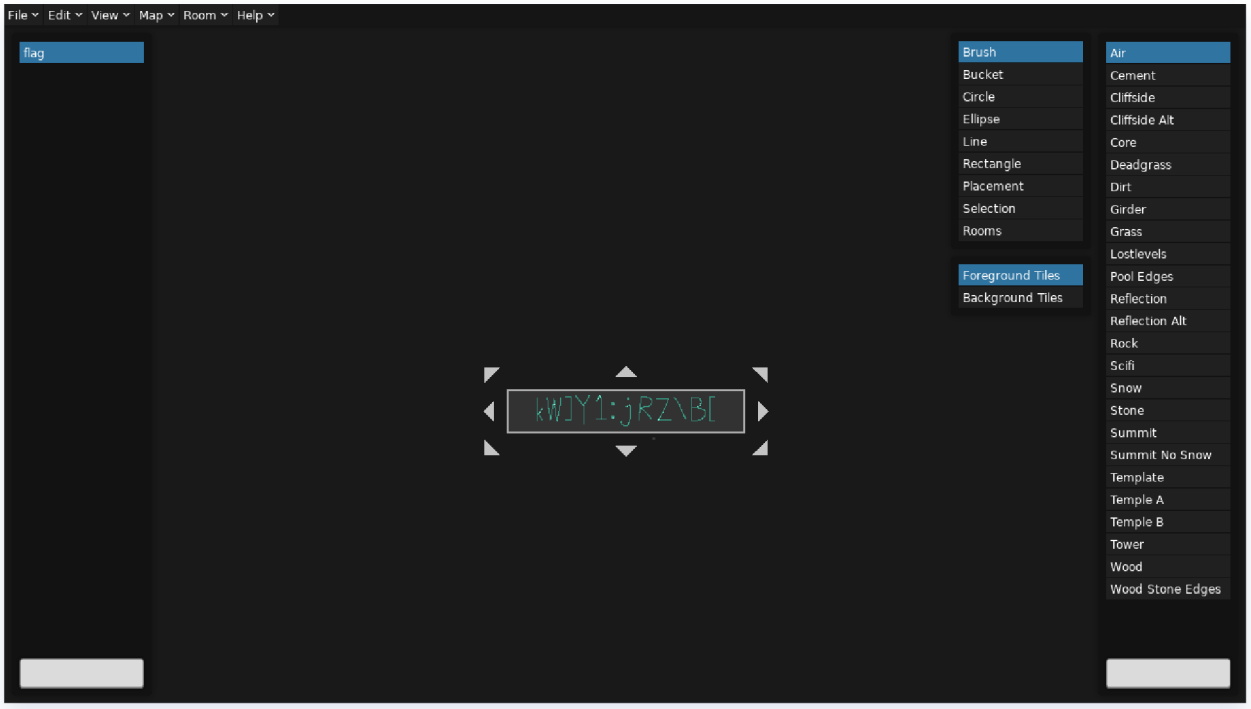




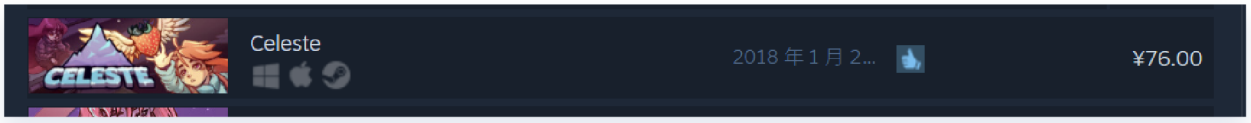
题目第四位有问题，用的是Q22，导致第四位得猜，我是根据异或之后文字的逻辑性试出来是n的，，，。。

拿到Loenn之后，根据文件名知道是和key.bin.xored进行异或，异或之后文件头是

CELESTE MAP，得知是蔚蓝游戏的地图，上网搜索到地图编辑器，但是没有这个游戏居然无法打开！！！！？？，蔚蓝给了多少钱？？？
问朋友借了一下steam账号，打开了：

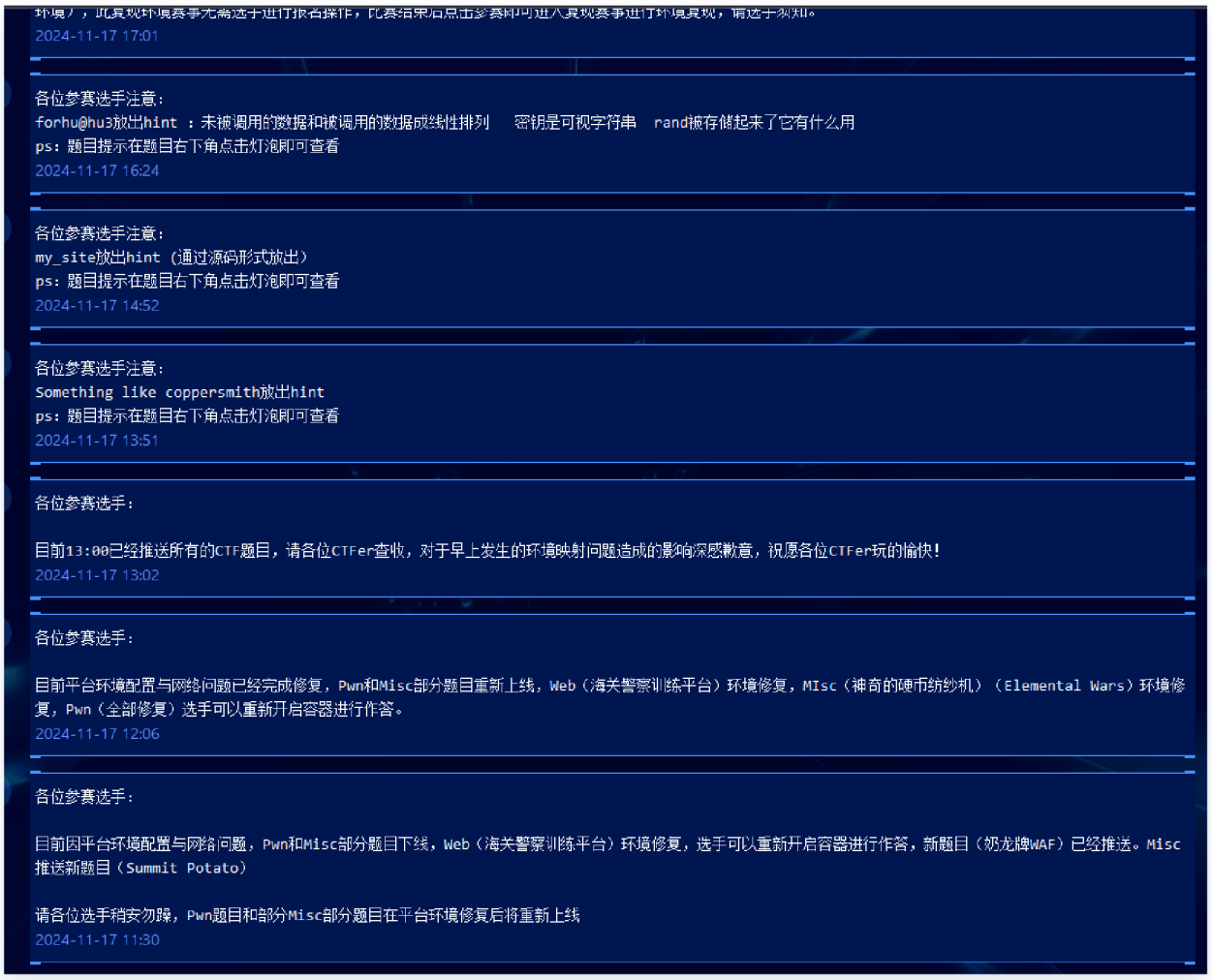


在xlsx改成zip后，media中有张图片，告诉我们和Zenith异或，把这个和Zenith异或之后拿到密码解压得到flag



不懂，做个题目要我花76去买游戏才能做吗？感觉有点幽默了

吐槽一下流量分析这个题目：



全程没在公告里看到有说这个题目附件换了，比赛信息和公告里也没看到群号，最后告诉我是在群里通知的，导致后面浪费了很久在错误的附件上，这点公告里也没有放出。。